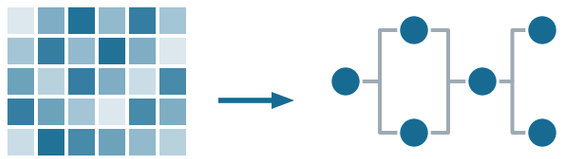
a Inputs and frame graph

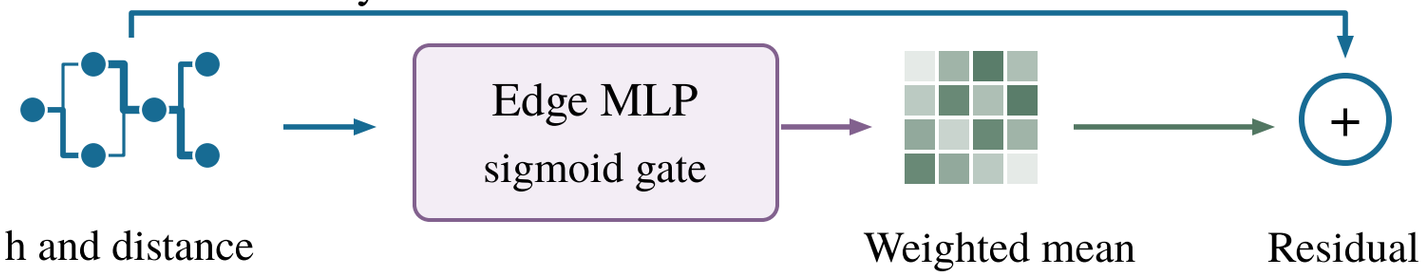


x : q ≤ 50

Within-frame E

No unrelated-field edges

b Gated residual layer



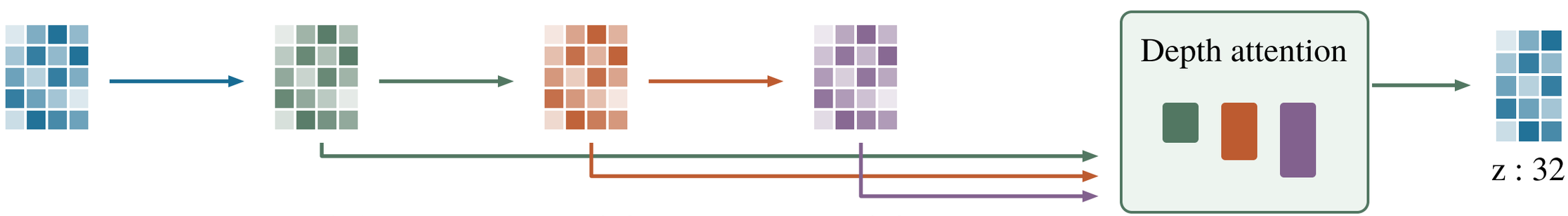
h and distance

Weighted mean

Residual

Normalize by gate sum → ReLU → dropout → LayerNorm

c Three depths → attention → embedding



Input

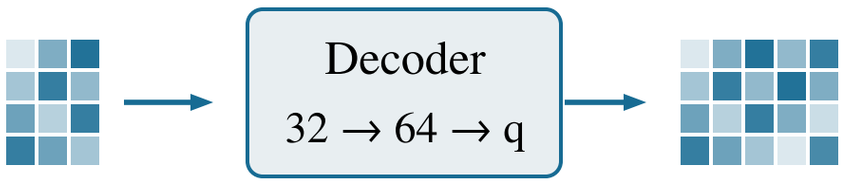
Depth 1

Depth 2

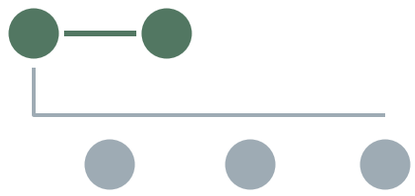
Depth 3

Three 64-dimensional hidden states

d Two training objectives



Reconstruction MSE

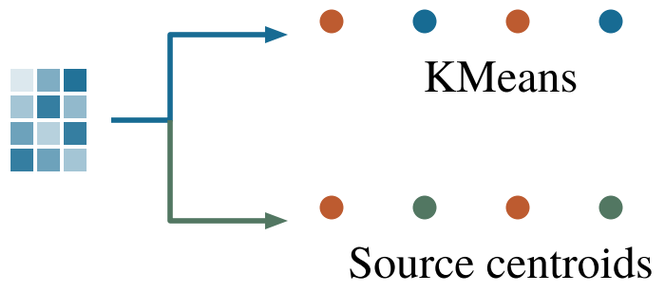


10 random negative draws

Detached gate weights; τ = 0.2

L = reconstruction + 0.5 × contrast

e Separate predictions



Seam score ≠ calibrated error